Practice Assignment 7.1: LangChain Tool Calling

In this exercise, you will learn how to extend an LLM with external capabilities using **LangChain Tool Calling**. You will configure a chat model, register tools (e.g., calculator and web search), and build a simple agent that decides when to call each tool. By the end, you'll observe how structured prompting and tool orchestration improve reliability and reduce hallucinations.

Step 1: Environment Setup

Ensure you have Python 3.9+ and install the required libraries:

- Install: `pip install langchain openai` (and a search provider like `google-search-results` if using SerpAPI).
- Set environment variables (e.g., `OPENAI_API_KEY, SERPAPI_API_KEY`) in your shell or notebook.

Step 2: Initialise Your LLM

Import the required classes and initialise a chat model (e.g., `ChatOpenAI`). Keep temperature low (e.g., 0.2) for predictable, tool-friendly outputs.

Step 3: Load Tools

- Register at least two tools: a calculator (e.g., `llm-math`) and a search tool (e.g., `serpapi`).
- Confirm each tool can be invoked independently before building the agent.

Step 4: Build the Agent

- Combine the chat model and tools using an agent initialiser (e.g., `initialize_agent`).
- Use a concise system prompt that instructs the model to call tools when needed and to return structured final answers.

Step 5: Run and Observe

- Issue queries that require math, web search, or both.
- Inspect intermediate reasoning (if available) and the tool outputs used to construct the final response.

- Record when the agent chooses the “wrong” tool or fails to call one, and adjust prompts accordingly.

Tasks

- **Task 1:** Configure LangChain with a chat model and two tools (calculator + search).
- **Task 2:** Write a query that requires only the calculator, e.g., “What is 512/16?”
- **Task 3:** Write a query that requires only the search tool, e.g., “Who is the current CEO of Microsoft?”
- **Task 4:** Write a combined query that forces both tools, e.g., “What is the square root of 81 plus the current US President’s first name?”
- **Task 5:** Reflect on how tool calling improves the reliability of LLM outputs.

4. Evaluation Checklist

- Did you set up the Python environment and initialise a chat model?
- Did you register and verify at least two tools (calculator and search)?
- Did your agent successfully call the calculator for a math-only query?
- Did your agent successfully call the search tool for a lookup query?
- Did your agent combine both tools for a multi-step query and return a coherent final answer?
- Did you reflect on tool selection, reasoning steps, and failure cases?

Note: *Working through the problems independently is key to building confidence and lasting understanding. This is a self-study activity for your practice and does not count towards programme completion. No submission is required. However, we highly recommend completing the activity on Vocareum to enhance your overall learning experience and build confidence in working with neural networks.*